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A Vision-Based Surveillance System for Livestock Health Monitoring Using Deep Learning

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Abstract—Livestock health plays a pivotal role in the sustainability and profitability of the agricultural sector. Early detection of diseases in animals such as cattle is critical to preventing outbreaks, reducing mortality, and ensuring animal

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[Linking Human Health and Livestock Health](#)

Mar 23, 2015 ♦ For most rural households in sub-Saharan Africa, healthy livestock play a key role in averting the burden associated with zoonotic diseases,♦...Missing: pivotal sus-

<https://pmc.ncbi.nlm.nih.gov/articles/PMC4370696>

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Feb 24, 2025 ♦ The early detection of a disease in a population of animals—a herd of cattle, for example—is particularly useful in controlling certain chronic♦...Missing: critical | Show results with:

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B. Behavior-Based Surveillance Systems

Beyond direct visual symptoms, livestock behavior has also been investigated as an indicator of animal health. In [6], the authors employed a YOLO-based object detection framework to track cattle movement and detect behavioral anomalies in real time. This approach enables early detection of stress, lethargy, or mobility issues, often preceding physical symptoms of disease. Combining behavioral surveillance with visual diagnostics enhances robustness and adds a non-invasive dimension to livestock health monitoring [7].

C. Use of CNN Architectures in Agriculture

Deep CNNs have seen growing adoption in precision agriculture tasks such as crop classification, pest detection, and livestock monitoring. Architectures like VGGNet, Inception, and ResNet have been evaluated for their ability to capture complex spatial patterns in image data. ResNet50, with its residual learning framework, has proven especially effective for fine-grained classification tasks due to its depth and gradi-

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Precision pest control and management in agriculture - GeoPard Tech

Firstly, cultural practices such as crop rotation, proper spacing, and maintaining healthy soil conditions help reduce pest populations. Secondly, biological...Missing: tasks classification,

<https://geopard.tech/blog/pest-management-in-precision-farming>

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Focal Scleral Nodule - EyeWiki

Mar 26, 2025 ⬢ These benign lesions are characteristically yellow or white, minimally elevated, round and postequatorial.Missing: swelling, essential cues

https://eyewiki.org/Focal_Scleral_Nodule

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Stay Balanced — The Importance of a Balanced Dataset in ML

A balanced dataset is one in which each class label has an approximately equal number of samples. For instance, if you're building a binary classifier to detect spam emails, a balanced dataset would contain roughly the same number of spam and non-spam...

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During experimentation, the model was trained in two phases: initially with the convolutional base frozen to train only the classifier head, and subsequently with all layers unfrozen to fine-tune the entire network. This approach helped in stabilizing early training and allowed the model to better adapt to domain-specific features in the latter stages [13]. Figure 5 displays the performance curves across epochs.

Fig. 4: Workflow of the proposed livestock health monitoring system

(a) Accuracy over epochs (b) Loss over epochs

Fig. 5: Training and validation accuracy and loss using ResNet50

VI. RESULTS

The performance of the proposed deep learning model was evaluated using the validation dataset consisting of cattle images labeled as either Healthy or Sick. The primary evaluation metric used was accuracy, calculated based on the number of correct predictions out of the total validation samples [13].

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[Performance of the proposed deep learning model for PLM prediction](#)

The performance of the proposed deep learning model was represented by the recall, precision, F1-score, and accuracy for the training and the test set. ...

https://www.researchgate.net/figure/Performance-of-the-proposed-deep-learning-model-for-PLM-prediction_tbl3_342627557

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[Do we automatically detect health- or general welfare-related issues ...](#)

The early detection of health disorders is a central goal in livestock production. Thus, a great demand for technologies enabling the automated detection of...Missing: correctly | Show results with:

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8113903>

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[Can data augmentation techniques compensate for the lack of data ...](#)

Reduced Overfitting**: By increasing the size of the training dataset, data augmentation can help to reduce overfitting, which can occur when the model is...Missing: absence affect

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